nature reviews cardiology

https://doi.org/10.1038/s41569-026-01253-3

Review article

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Molecular damage associated with ageing drives inflammation in cardiovascular disease

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Abstract

Chronic inflammation has long been recognized as a major risk factor for and a causal contributor to cardiovascular disease (CVD). However, advances in omics technologies and deepening insights into CVD pathogenesis have expanded our understanding of the underlying mechanisms. Inflammation is now seen not as an isolated cause, but as one of several biological responses to cumulative tissue damage over time. In this Review, we propose that inflammation initially functions as a resilience mechanism, acting to resolve molecular and cellular damage driven by environmental stressors and intrinsic age-related entropy. With ageing, however, this protective response can become dysregulated and maladaptive, promoting collateral pathological changes. We illustrate this theory through two examples, atherosclerosis and age-related impairment of tissue perfusion, and support these conceptual models using proteomic data from large population studies with cardiovascular outcomes. Our findings reaffirm the central role of inflammation in CVD pathophysiology, but also indicate that the upstream biological driver of inflammation is molecular damage that is either not readily prevented or repaired by inadequate resilience mechanisms. Understanding the coordination of these responses offers new opportunities for targeted prevention and treatment of CVD.

Sections

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Key points

- Ageing, endothelial dysfunction and impaired perfusion lead to molecular damage and trigger resilience mechanisms to maintain tissue integrity.
- Persistent, unrepaired molecular damage activates inflammatory responses that, over time, contribute to atherosclerosis.
- In ageing, impaired vascular regulation and tissue fibrosis can cause recurrent hypoperfusion, leading to energy deficits and progressive molecular damage.
- Chronic inflammation acts as a component of stress response pathways that, ultimately, lead to the onset and progression of cardiovascular disease.
- Biomarkers of cardiovascular disease, such as fibroblast growth factor 23 and growth/differentiation factor 15, reflect not only inflammation, but also changes in the extracellular matrix, cellular senescence and proteostasis.

Introduction

Inflammation has emerged as a crucial factor in the development of cardiovascular disease (CVD). Extensive research on the underlying molecular mechanisms, characterization of structural changes within the cardiovascular system, examination of morbidity and mortality in the general population, and outcomes from clinical trials highlight the crucial role of inflammation in heart and vascular pathology. Key biomarkers, such as C-reactive protein (high-sensitivity assay) and IL-6, were initially thought to be exclusively produced by immune cells or induced during inflammatory conditions, leading to the hypothesis that immune-based inflammation is a fundamental mechanism in the aetiology of CVD[1,2]. Moreover, circulating levels of these biomarkers tend to increase with age, a phenomenon referred to as 'inflammageing'. This measurable outcome has been attributed to intrinsic dysregulation of the immune system and immunosenescence, contributing to the rising incidence of CVD with increased age[3]. Notably, the CANTOS trial[4] demonstrated that inhibiting the pro-inflammatory cytokine IL-1β with the monoclonal antibody canakinumab effectively reduces the risk of acute cardiovascular events. This finding established the causal link between inflammation and CVD, but the underlying mechanisms remained poorly defined, and there is an urgent need for cardiovascular risk models that incorporate novel, mechanism-based risk markers[5].

Technological advances over the past decade have enabled the simultaneous measurement of thousands of proteins and metabolites in small blood samples[6]. Findings from large epidemiological studies using these cutting-edge platforms are challenging the notion that increased levels of pro-inflammatory biomarkers result solely from intrinsic immune dysfunction (discussed in detail in the section 'Inflammation as a biomarker of CVD risk'). Instead, the accumulation of macromolecular damage arising from environmental insults or ageing can trigger a resilience response that combines an inflammatory component with other stress-response mechanisms in somatic tissues[7].

In this Review, we posit that intrinsic or induced macromolecular damage constitutes a stress insult that elicits a protective resilience response. However, if this stress–resilience response is not balanced, persistent damage will accumulate and trigger inflammation, a defensive process that can become maladaptive over time and contribute to pathology. In support of this concept, we theorize that growth/differentiation factor 15 (GDF15), which is a member of the transforming growth factor β (TGFβ) superfamily and a core protein of the senescence-associated secretory phenotype (SASP)[8], is likely to reflect an adaptive response to mitochondrial stress rather than being a direct indicator of inflammation. Other markers, such as matrilysin (also known as matrix metalloproteinase 7 (MMP7)), macrophage metalloelastase (also known as MMP12) and renin – enzymes involved primarily in extracellular matrix (ECM) breakdown and tissue signalling – exhibit pleiotropic biological activities that are only partially related to inflammation and are regulated by non-immune tissues. Therefore, we propose that the rising levels of inflammatory biomarkers in blood and tissues, seen both with increasing age and in CVD, is the collective consequence of resilience mechanisms in response to cellular stress[9], including, but not limited to, the classic immune response.

By critically evaluating the literature, we support our hypothesis using two examples relevant to CVD, namely atherosclerosis and chronic tissue hypoperfusion due to microvascular dysregulation. The latter results in reduced tissue oxygenation and increased resistance to energy flow through the mitochondria[7], a metabolic state that activates a robust cellular stress response characterized by the release of pro-inflammatory markers. Furthermore, we summarize published findings indicating that a wide array of circulating blood biomarkers serve as risk indicators for CVD, with many but not all indicating immune system involvement. We propose a comprehensive biomarker profile, which includes links to pathways such as proteostasis, as well as molecular and architectural changes in the ECM. We believe that this profile can enhance our understanding of the mechanisms underlying CVD and inform targeted therapeutic strategies to mitigate the adverse effects of inflammation, while preserving the essential protective functions of cytokine signalling.

Molecular damage and inflammation

Ageing and entropy

Ageing is by far the strongest risk factor for CVD. One prominent model of ageing is based on the second law of thermodynamics, which posits that the total entropy of an isolated system never decreases[10]. Although organisms are not closed systems and their entropy can transiently decrease through molecular repair processes, over time organisms tend to progress towards higher entropy and irreversible accumulation of stochastic damage. This process can be accelerated by various internal and external stressors, resulting in defective or maladaptive responses that no longer maintain homeostasis.

Resilience responses

To combat this inevitable trajectory towards increased entropy, highly efficient resilience mechanisms tailored to specific molecules and types of damage have evolved to maintain the integrity and function of the organism[11] (Fig. 1). As organisms exist as 'social' cell collectives[12], a protective response to stress activated by a single cell must be shared by other cells to mobilize an efficient, integrated response. Therefore, many resilience mechanisms are tied to the production of signalling molecules, such as cytokines. Triggered by molecular damage, cytokine release is used as a signalling strategy to warn other cells about a challenge or threat. This model is consistent with the idea that inflammation has evolved as a response to acute stressors, such as infections or trauma, to restore physiological homeostasis to the organism[9]. Below,

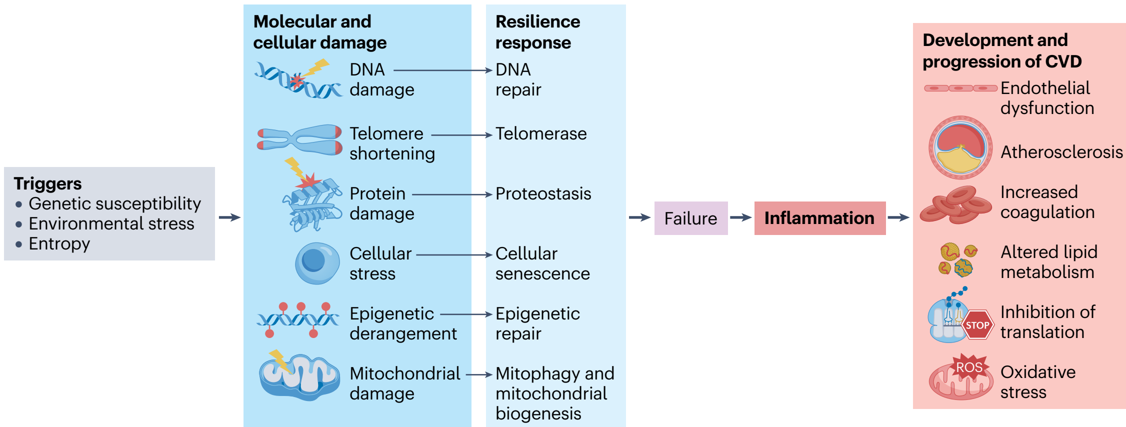

Fig. 1 | Resilience mechanisms that counter ageing-related cellular insults. Over time, the accumulation of cellular damage from various triggers, including genetic alterations, environmental stress and entropy, leads to increased inflammation, which contributes to the development and progression of cardiovascular disease (CVD) phenotypes through several biological mechanisms. ROS, reactive oxygen species.

we present three generalized examples of resilience responses at the molecular and organelle level.

DNA. DNA damage and DNA repair mechanisms serve as a prime example of the interplay between stressors and resilience response mechanisms. Unlike proteins or RNA, DNA does not undergo turnover. Unrepaired genomic damage can, therefore, persist for the lifespan of the cell and can be propagated to daughter cells. DNA in human cells is subject to continuous damage (estimated at 10,000–100,000 events per day), primarily due to spontaneous decay, modification by endogenous reactive chemicals, replication errors and environmental influences[13]. Somatic mutations accumulate over time, and persistent DNA lesions can trigger inflammation through interconnected pathways[14,15]. For example, genomic stress activates a DNA damage response[16] that upregulates cellular tumour antigen p53, which subsequently induces the expression of pro-inflammatory cytokines, such as IL-1β, IL-6 and tumour necrosis factor (TNF)[17].

Despite effective DNA repair mechanisms, severe and irreparable DNA damage can trigger persistent cell cycle arrest (cellular senescence) or programmed cell death (apoptosis), thereby preventing potential harm to the organism, such as mutation-driven neoplastic transformation. Damage-induced senescent cells enter a stable state of growth arrest, and if they are not cleared by the immune system, they perpetually secrete a variety of bioactive molecules (collectively termed SASP) that act on surrounding tissues and recruit immune cells, thereby perpetuating chronic inflammation[18]. In addition, nuclear DNA released from apoptotic or necrotic cells can be sensed by the immune system as a danger signal, further amplifying the inflammatory response.

Proteins. Protein damage and relevant surveillance mechanisms provide another example of the resilience response to stress. Protein function relies on precise amino acid sequence and polypeptide folding. The unfolded protein response (UPR) increases the expression of molecular chaperones to correct errors in folding and promotes the degradation of severely misfolded proteins[19]. Activation of the UPR, which is mainly driven by the accumulation of unfolded or misfolded proteins in the lumen of the endoplasmic reticulum, leads to the production of pro-inflammatory cytokines, such as IL-1β, IL-6 and TNF, and the activation of the nuclear factor κB (NF-κB) pathway.

Mitochondria. As organismal entropy increases, damage to individual molecules translates to organelles and, eventually, whole cells. At the organelle level, mitochondrial quality control is a primary example of a resilience response. This process eliminates dysfunctional mitochondria through various forms of mitophagy, tightly coupled to mitochondrial biogenesis[20]. If dysfunctional mitochondria are not removed, they can create an oxidative environment via the release of reactive oxygen species (ROS), as well as liberate oxidized mitochondrial DNA (mtDNA) and oxidized cardiolipin. The immune system recognizes these products as danger-associated molecular patterns (DAMPs), triggering an inflammatory response[6]. Dysfunctional mitochondrial quality control also drives a retrograde (mitonuclear) signalling programme that impairs cellular identity in several cell types[21].

Resilience responses in CVD

Under conditions of chronic cellular stress, resilience response triggers can become unchecked, with maladaptive and persistent activation of the associated mechanisms. This process can lead to the accumulation of damage and the accelerated progression of pathological conditions, such as CVD. In the following sections, we explore the molecular, biological and pathophysiological interactions between the accumulation of molecular damage, resilience mechanisms, cellular senescence and inflammation in atherosclerosis[22] and in dysregulated microvascular tissue perfusion[23].

Atherosclerosis–inflammation–atherosclerosis

Although, historically, two distinct theories underlying the development of atherosclerosis were postulated (the lipid hypothesis and the inflammation hypothesis), the current view entails a unified lipid-driven inflammatory disease paradigm. In this integrated theory, LDL particles are postulated to attach to scavenger receptors on the surface of arterial endothelial cells and transported to the subendothelial space where they are oxidized by enzymes, such as myeloperoxidase, or by ROS to form oxidized LDL (oxLDL). Macrophages then absorb oxLDL and turn them into foam cells, which release inflammatory and chemotactic molecules.

These signals attract more immune cells, enhance inflammation in the arterial wall, cause smooth muscle cells to lose their normal function and, eventually, lead to plaque accumulation[24]. This model of plaque development supports the notion that inflammation is not the sole, primary mechanism, but rather a response to an initial molecular insult[25]. In this Review, we focus on the pivotal role of non-immune cells in merging the lipid-based and inflammation-based theories. Specifically, we explore how prolonged or excessive activation of resilience mechanisms, such as senescence, contribute to the initiation of atherosclerosis.

Initial damage and endothelial response. In response to various stresses and inflammation, a cascade of maladaptive processes develops within the endothelium, contributing to the progression of atherosclerosis (Fig. 2). Endothelial cells form a monolayer that lines the arterial lumen, serving as the primary barrier between blood and surrounding tissues. These cells have a crucial role in regulating vascular tone, coagulation, angiogenesis and inflammation. Impaired endothelial cell metabolism can lead to molecular damage and activate signalling pathways that trigger an immune response. Endothelial cells are also particularly vulnerable to a range of circulatory factors that can cause damage and provoke resilience responses. Key stressors include turbulent blood flow, hypercholesterolaemia, hyperuricaemia, diabetes mellitus and hypertension[26,27]. Experimental studies have also implicated oxidative stress, alterations in calcium levels, advanced glycation end products, mitogens and mitochondrial damage as causative mechanisms in the development of atherosclerosis[28].

The fate of endothelial cells following exposure to external insults is largely determined by the severity of the damage. Severe DNA damage that exceeds the cell's capacity for repair, for example, can activate apoptosis or alternative forms of cell death. Conversely, if the damage is

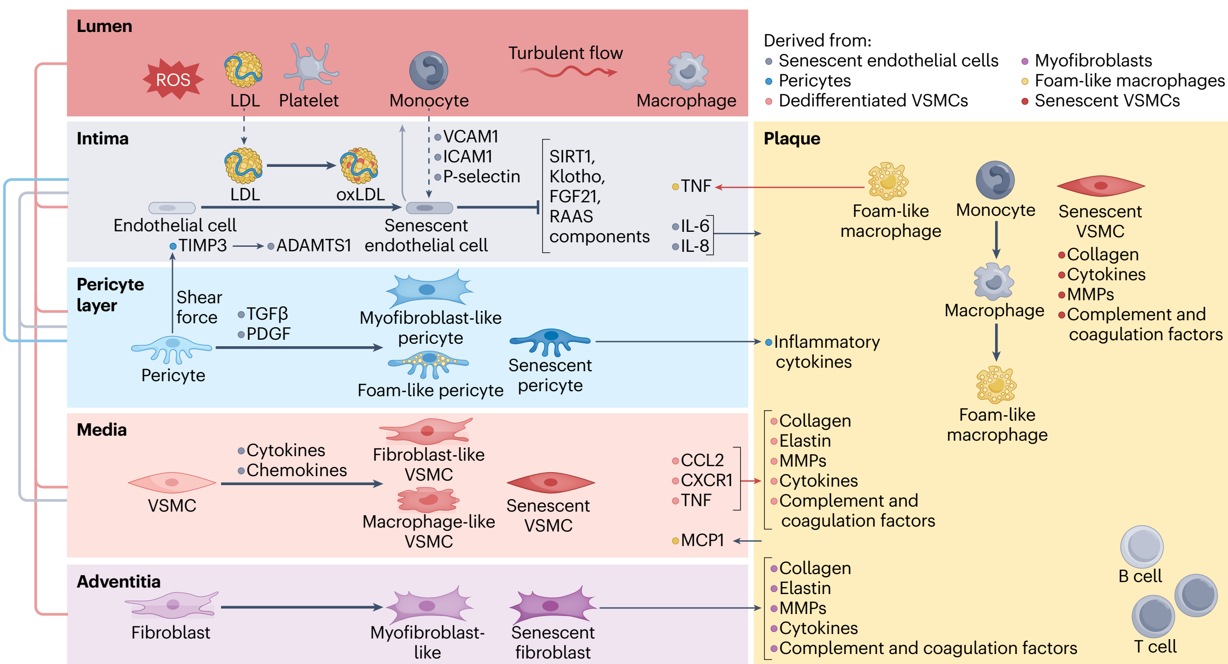

Fig. 2 | Coordinated stress and inflammatory signalling network during atherosclerosis. The complex architecture of the arterial wall, illustrating the flow of second messenger signals between cells and arterial layers that drive phenotypic and functional changes. Stress and damage within the lumen caused by stimuli such, as reactive oxygen species (ROS) and turbulent flow, promote the entry of LDL particles into the subendothelial space, where they become oxidized. This event initiates a cascade of stress-response pathways leading to inflammatory plaque formation. In the intima, vascular endothelial cells shift from a quiescent to a senescent or activated state, resulting in increased production of inflammatory mediators, such as intercellular adhesion molecule 1 (ICAM1), P-selectin and vascular cell adhesion protein 1 (VCAM1). These molecules promote monocyte adhesion and infiltration, reinforcing plaque development. Senescent endothelial cells also suppress protective signalling molecules (such as fibroblast growth factor 21 (FGF21), klotho, components of the renin–angiotensin–aldosterone system (RAAS) and NAD-dependent protein deacetylase sirtuin-1 (SIRT1)) while elevating IL-6 and IL-8 secretion into the plaque environment. In the pericyte layer, pericytes respond to shear stress and can transform into myofibroblast-like, foam-like or senescent cells through stimulation by platelet-derived growth factor (PDGF) or transforming growth factor-β (TGFβ). These altered pericytes release inflammatory cytokines that further enhance plaque growth. In the media, quiescent vascular smooth muscle cells (VSMCs) react to local cytokines and chemokines by dedifferentiating into fibroblast-like, macrophage-like or senescent phenotypes. These pathological VSMCs secrete C-X-C chemokine receptor type 1 (CXCR1), C-C motif chemokine 2 (CCL2) and tumour necrosis factor (TNF), amplifying vascular inflammation and disease progression. The adventitia, composed of fibroblasts and connective tissue, is less well characterized than the other layers of the vessel wall. However, under stress, fibroblasts can also adopt myofibroblast-like or senescent features. Myofibroblast-like cells produce collagen, extracellular matrix components and cytokines that contribute to plaque formation. The resulting plaque comprises a complex mixture of immune and vascular cells, including foam-like macrophages and senescent VSMCs. These cells perpetuate inflammation and lipid accumulation through their secreted factors, progressively altering the arterial wall (red arrows). Overall, the figure illustrates that inflammation arises as a consequence of accumulated cellular stress and damage, processes that intensify with ageing and disease. Blunt-end lines represent inhibition or reduction and arrows indicate increased signalling activity. ADAMTS1, a disintegrin and metalloproteinase with thrombospondin motifs 1; MCP1, monocyte chemoattractant protein-1; MMPs, matrix metalloproteinases; oxLDL, oxidized LDL; TIMP3, metalloproteinase inhibitor 3.

sublethal, cells can become senescent. The manifestation of senescence varies by cell type and the specific triggers involved. In endothelial cells, one of the most notable effects of senescence is the upregulation of adhesion molecules, such as vascular cell adhesion protein 1 and intercellular adhesion molecule 1 (ICAM1)[29,30], making the cells 'sticky' and promoting the engagement of circulating leukocytes, primarily monocytes. Once monocytes migrate into the vascular wall, they differentiate into macrophages and engulf oxLDL, which cannot be processed, leading to their conversion to lipid-laden foam cells. These foam cells worsen the inflammatory response and contribute to atherosclerosis by releasing cytokines, such as TNF, which promote further upregulation of adhesion molecules on endothelial cells. In addition, foam cells release C-C motif chemokine 2 (CCL2; also known as monocyte chemoattractant protein 1), a factor that stimulates the proliferation of vascular smooth muscle cells (VSMCs) and attracts more leukocytes to the intimal space. This cascade of events leads to the formation of fatty streaks and, subsequently, the development of plaques[31,32].

Senescent endothelial cells also propagate inflammation directly by secreting SASP factors, such as IL-1β, IL-6 and the chemokine C-X-C motif chemokine (CXCL) 11 (refs. 33,34). In addition, senescent endothelial cells upregulate classic cell cycle arrest pathways, such as p53–cyclin-dependent kinase inhibitor 1 (also known as p21), leading to reduced nitric oxide (NO) production, impaired regenerative capacity and loss of angiogenic properties[28]. Aside from the production of inflammatory markers, senescent endothelial cells upregulate a range of stress response pathways, such as mitogen-activated protein kinase, serine/threonine-protein kinase mTOR, the UPR, NF-κB and other major pro-inflammatory mechanisms, leading to a further increase in the SASP and overexpression of ECM remodelling proteins that promote plaque instability. Moreover, protective factors, such as NAD-dependent histone deacetylase sirtuin-1, Klotho, fibroblast growth factor 21 (FGF21) and components of the renin–angiotensin–aldosterone system (RAAS) are downregulated in the presence of senescent endothelial cells. However, the role of these changes in atherosclerosis is not clear[27]. Increased stress on endothelial cells also influences telomere integrity, reducing telomerase activity and increasing ROS and nitrosative stress, promoting further local cellular senescence[30].

These biological responses might initially be beneficial. For example, the RAAS can activate the angiotensin II type 2 receptor and favourably modulate inflammation and fibrosis. However, prolonged activation of the RAAS, due to inefficient resolution, can increase the expression of the angiotensin II type 1 receptor, elevating ROS production, inflammation and ECM remodelling, and thereby exacerbate CVD by both accelerating the atherosclerotic process and negatively affecting blood flow haemodynamics[35]. Overall, endothelial cellular senescence and maladaptive stress responses have been linked to imbalanced vascular tone, disruption of endothelial cell tight junctions, increased vascular permeability, and impaired angiogenesis and mitochondrial biogenesis, all of which contribute to the activation and dysfunction of the immune response and to the progression of atherosclerosis.

Second-level reaction and exacerbation by VSMCs. Healthy VSMCs form a multicellular layer in the arterial wall, architecturally supported by elastic fibres that help to regulate blood pressure and flow through constriction and relaxation. In the early stages of atherosclerosis, particularly in the presence of senescent endothelial cells, excessive ROS and oxLDL can cause DNA damage and trigger VSMC senescence (Fig. 2). Senescent VSMCs secrete typical SASP factors, such as IL-1β and IL-6, as well as complement and coagulation factors, which contribute to plaque vulnerability and perpetuate inflammation[36,37]. Elevated levels of inflammatory cytokines, chemokines, growth factors and stress response molecules (such as platelet-derived growth factor (PDGF)-BB and TNF) within the subendothelial space promote VSMC dedifferentiation. During this process, these normally contractile cells lose their specialized function and adopt a more proliferative and migratory state that contributes to plaque development[38]. This process reduces the ability of VSMCs to regulate blood flow, prevent clot formation, and to support molecular exchange between the blood and surrounding tissues. Fibroblast-like VSMCs, a specific form of VSMC dedifferentiation, secrete ECM remodelling proteins, such as MMPs, collagen and elastin. These factors enhance cell–matrix reorganization and further stimulate VSMC proliferation. Macrophage-like VSMCs, another dedifferentiated form of these cells, acquire foam cell-like properties and contribute to the recruitment and activation of leukocytes, increasing the risk of plaque rupture and acute ischaemic events. Consistent with the lipid-driven inflammation theory, macrophage-like VSMCs release inflammatory mediators, such as CCL2, C-X-C chemokine receptor type 1 (CXCR1) and soluble TNF[39]. Importantly, maintaining the VSMC contractile phenotype and preventing their dedifferentiation has been shown to reduce the severity of atherosclerosis, underscoring the crucial role of these cells in vascular disease[40].

The interplay between highly heterogeneous VSMCs, including the synthetic and senescent forms, contributes to the formation of the atherosclerotic plaque cap. The stability of this cap is crucial for preventing plaque rupture, arterial thrombotic occlusion and acute ischaemia in the affected territory. An unstable cap is characterized by a thin fibrous layer of VSMCs, heightened inflammatory cell infiltration and MMPs that degrade the ECM. By contrast, a stable plaque is marked by a thick fibrous VSMC cap, largely free of immune cells and enriched with collagen, among other protective features[41]. The stress response pathways that influence VSMC fate, promoting either ECM deposition or degradation as well as inflammation, are important factors that affect the severity of vascular disease and the risk of atherosclerosis. The majority of acute coronary syndrome events result from the sudden rupture of an atherosclerotic plaque, rather than from a gradual narrowing of the artery[42]. Therefore, the identification of vulnerable plaques in asymptomatic individuals is crucial.

Several biomarkers are associated with thin-cap fibroatheromas, such as increases in circulating levels of amyloid A, fibrinogen, homocysteine, high-sensitivity C-reactive protein, IL-6 and plasminogen activator inhibitor 1, as well as reductions in levels of HDL and bile acids[43]. These biomarkers implicate inflammation as well as other processes, including fibrosis, lipid deposition, proteostasis, and tissue healing and regeneration in maintaining cap stability. MMP9, pentraxin-related protein PTX3 and TNF have also been associated with disease severity and plaque vulnerability[44]. Among their other physiological functions, these molecules are members of the SASP, indicating that cellular senescence is involved in plaque instability[44]. In addition, an elevated circulating level of FGF23 has been proposed as a potential indicator of high-risk CVD and cardiovascular mortality[45] and, in 2020, angiopoietin-related protein 3 was reported to be a potential biomarker of major adverse cardiovascular events in patients with coronary artery disease[46]. Although the quest for specific and insightful biomarkers for vulnerable plaques and severe atherosclerotic disease is ongoing, current evidence suggests that growth factors, ECM-related proteins and metabolic biomarkers could offer valuable insights alongside the existing array of circulating inflammatory proteins.

The integration of stromal cells. Often overshadowed by the importance of VSMCs, fibroblasts and pericytes also contribute to atherosclerosis (Fig. 2). In addition to maintaining the structural integrity of arteries by secreting collagen and ECM-modifying proteins, fibroblasts respond to prolonged stress by secreting cytokines, growth factors and ECM proteins, and can transition into myofibroblasts to participate in vessel remodelling or plaque development[47]. Arterial fibroblasts are activated by ischaemia–reperfusion injury, direct tissue damage, inflammation and ROS, triggering the release of PDGF, TGFβ and oestrogens, which facilitate the fibroblast-to-myofibroblast transition[48]. Research has shown that blocking PDGF receptor β and its ligand PDGF-BB inhibits neointimal formation in rat carotid arteries[49]. This process is particularly relevant because increased numbers of myofibroblasts and adventitial remodelling have been observed to precede intimal and medial remodelling in porcine carotid arteries subjected to hypercholesterolaemia[50]. In addition, activated fibroblasts in the pulmonary artery secrete various factors, including inflammatory cytokines and chemotactic proteins, such as CCL2, as well as matricellular proteins that facilitate leukocyte recruitment. However, the specific roles of fibroblasts in attracting leukocytes, and in the broader inflammatory state in atherosclerosis, remain incompletely understood[51]. Myofibroblasts from human atherosclerotic plaques exhibit enrichment of immune cell-specific transcripts, such as *CD74* and *CD4* mRNAs, underscoring their role in stress-induced inflammation during CVD and highlighting their heterogeneity[52]. Despite their prevalence and importance in the adventitia, understanding the molecular implications of myofibroblasts and potentially senescent fibroblasts has been challenging owing to difficulties in specific labelling and genetic manipulation[47].

Pericytes are highly heterogeneous mural cells that contribute to the integrity of the vascular network through their close association with endothelial cells[53]. Once thought to be exclusive to the microvasculature, pericyte or pericyte-like cells have now been found in the walls of larger blood vessels[54]. Our current understanding of the role of pericytes in atherosclerosis is limited. We know that prolonged activation of stress response pathways associated with TGFβ and PDGF induces a transition from pericytes to myofibroblasts and the detachment of cells with migratory and fibroblast-like properties from the arterial wall[55]. In response to shear stress, pericytes upregulate metalloproteinase inhibitor 3 to counteract elevated levels of a disintegrin and metalloproteinase with thrombospondin motifs 1 in damaged endothelial cells[56]. This finding suggests that pericytes help prevent ECM degradation and maintain vascular integrity under conditions of stress[56]. Furthermore, pericytes have been shown to engulf modified LDL, transforming into foam cells and secreting inflammatory cytokines that recruit and retain immune cells within the atherosclerotic plaque[53,57]. In the section below on 'Mechanisms that regulate tissue perfusion', we further explore the multifaceted role of pericytes, highlighting their function as environmental sensors and transducers of stress signals, thereby contributing to signalling cascades that can lead to both physiological and pathological outcomes.

The effects of ageing. Biological damage caused by ageing and other harmful factors challenges the resilience mechanisms that maintain cell integrity and the function of arterial wall components. This damage triggers stress-response mechanisms that activate an inflammatory reaction, which initially serves a protective and compensatory purpose. Over time, however, the persistent activation of these stress-response mechanisms, including inflammation, contributes to the progression of pathology and accelerates the atherogenic process (Fig. 2). Therefore, although managing inflammation can have beneficial effects on CVD, identifying strategies to reduce the accumulation of damage responsible for the maladaptive stress response and chronic inflammation is equally, if not more, crucial. Importantly, the events leading to the formation of atherosclerotic plaques involve both immune and non-immune cells. Although we focus on non-immune types in this Review, immune cells are also affected by ageing and the various forms of stress-induced cellular damage that contribute to inflammation during atherosclerosis[58].

Chronic microvascular tissue hypoperfusion

Obstructions in perfusion, such as those caused by atherosclerosis, can lead to acute and severe health crises. The myocardium utilizes 65–75% of the oxygen from coronary blood flow, which renders the heart particularly vulnerable to even minor reductions in oxygen delivery, resulting in persistent damage when ischaemia is prolonged. Energetically, hypoxia takes away the main electron acceptor within mitochondria, notably increasing energy resistance[7]. The mechanisms underlying myocardial damage during prolonged ischaemia, as well as the resilience mechanisms that aim to minimize and repair damage during hypoxia and ischaemia–reperfusion injury, have been investigated in numerous studies with an emphasis on the acute inflammatory response. In acute myocardial infarction, inflammatory processes can exacerbate necrosis, while simultaneously clearing dead cells, producing growth factors and promoting angiogenesis, all of which contribute to myocardial repair and healing[59]. However, excessive inflammation can lead to increased myocardial fibrosis, resulting in heightened cardiac stiffness, reduced functional capacity and an increased risk of arrhythmias.

The inflammatory processes associated with acute myocardial infarction have been extensively studied and will not be the focus of our Review. Instead, we concentrate on chronic tissue hypoperfusion, which is often paucisymptomatic and, with the exception of chronic brain hypoperfusion[60], has garnered comparatively little attention. To explore how even minor decreases in oxygen delivery might contribute to stress responses and inflammation, we outline the mechanisms that ensure sufficient provision of oxygen and nutrients in response to local and global fluctuations in demand. A full understanding of these physiological processes can provide valuable insights into the mechanisms underlying chronic inflammation[60].

Mechanisms that regulate tissue perfusion. Optimal tissue perfusion involves the complex interplay between various vascular structures and the processes that control blood flow (Fig. 3). Large-sized and medium-sized arteries carry oxygenated blood from the heart to arterioles, regulating blood flow to tissues. Arteries can constrict or dilate to adjust their resistance, affecting blood pressure and perfusion to downstream capillaries. Capillaries facilitate the exchange of gases, nutrients and waste products between blood and tissues.

The regulation of tissue perfusion in response to metabolic demands operates at various levels within this network. The contraction or relaxation of smooth muscle within arteriole walls is regulated by the sympathetic nervous system (vasoconstriction), as well as circulating hormones with vasoconstrictive properties, such as angiotensin II, adrenaline, noradrenaline and vasopressin[61]. Conversely, vasodilatory factors, such as atrial natriuretic peptide, insulin, NO, prostacyclin and prostaglandin E_2, have important roles in smooth muscle relaxation. When tissues are active and demand for oxygen

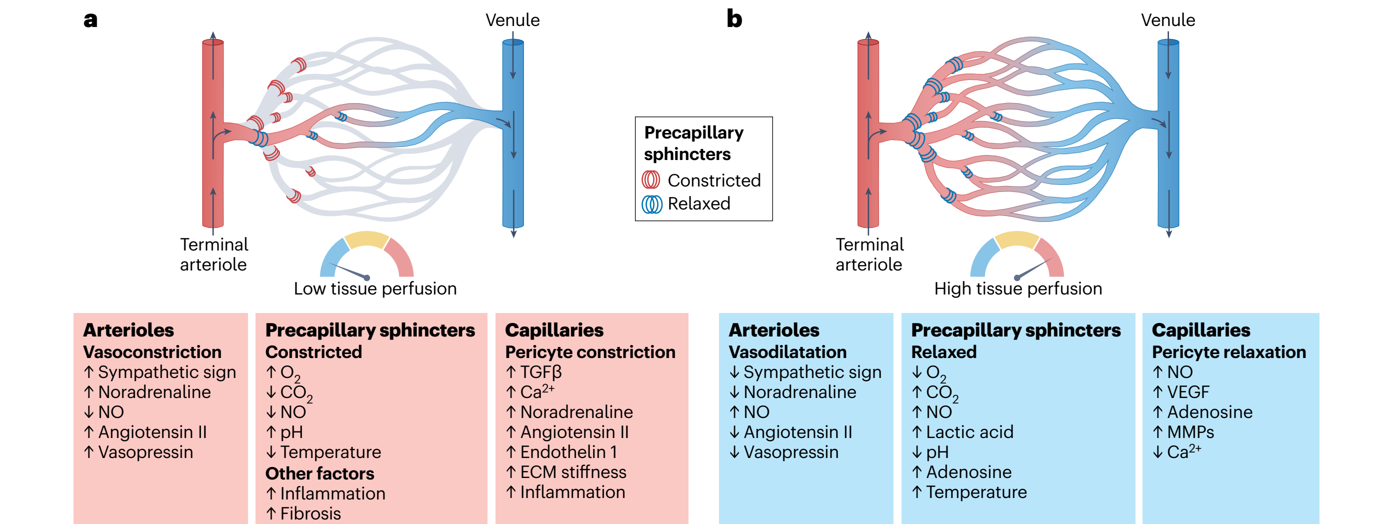

Fig. 3 | Factors and mechanisms that modulate vascular regulation of tissue perfusion. a, Constricted precapillary sphincters limit capillary blood flow leading to tissue hypoperfusion. **b**, Relaxed precapillary sphincters allow unrestricted blood flow, enhancing tissue perfusion. The information below each illustration shows the effects of various stimuli on tissue perfusion: ↑ indicates an increased effect and ↓ represents a decreased effect. The multiplicity of regulatory mechanisms fine tunes local tissue perfusion to meet metabolic demands. ECM, extracellular matrix; MMPs, matrix metalloproteinases; NO, nitric oxide; TGFβ, transforming growth factor β; VEGF, vascular endothelial growth factor.

increases, NO is synthesized from L-arginine by endothelial nitric oxide synthase (eNOS) and diffuses into nearby smooth muscle cells. There, it activates guanylate cyclase, leading to the production of cyclic GMP, which causes smooth muscle relaxation, resulting in vasodilation and increased blood flow. By dilating arterioles and enhancing blood flow, NO facilitates capillary recruitment and improves the delivery of oxygen and nutrients to the tissues. Precapillary arterioles actively distribute blood throughout the body, ensuring that areas with high physiological activity receive adequate perfusion. Their junctions with capillaries are controlled by local factors, such as oxygen and carbon dioxide levels, pH levels and metabolic by-products[62]. For example, increased carbon dioxide and NO and decreased oxygen levels in tissue can signal to VSMCs to relax, allowing more blood to flow into that region[63].

The number, density, haematocrit and permeability of capillaries are vital factors influencing tissue perfusion. Despite comprising <5% of total blood volume, the capillary system shows substantial adaptability to the fluctuation in oxygen demand in the tissue. In some tissues, precapillary sphincters at the junction between arterioles and capillaries can constrict or relax to control blood flow, and selectively open or close capillaries based on the metabolic needs of the tissue[64]. During periods of increased metabolic activity, the velocity and flux of blood, as well as haematocrit, vary substantially across the capillary network, allowing regulation of oxygen diffusion to tissues[65]. Similarly, during food absorption, blood flow is strategically redistributed to the digestive tract with reduced perfusion of peripheral areas, such as muscles and skin. Within the capillaries, blood flow is regulated by pericytes, which are specialized contractile cells embedded in the basal membrane of these vessels and in post-capillary venules[66]. These elongated or stellate cells are found in various tissues, but are particularly abundant in the central nervous system, where they maintain the integrity of the blood–brain barrier[67]. Pericytes contract in the presence of actin and other cytoskeletal proteins within the basal membrane[66], thereby providing structural support, while also regulating blood flow and capillary permeability. Emerging evidence suggests that pericytes can also respond to injury and to the presence of cytokines, contributing to tissue repair and remodelling[68].

Dysregulation of tissue perfusion. The interconnected mechanisms that regulate tissue perfusion in response to metabolic and oxygen demands often become dysfunctional with ageing and various disease states[69]. Under normal physiological circumstances, oxygen delivery substantially exceeds oxygen consumption, enabling electrons extracted from hydrocarbons in the Krebs cycle to flow with the least resistance towards oxygen in the mitochondria. When oxygen delivery is limited, electrical resistance increases, causing a backflow of electrons onto the Krebs cycle electron carrier NAD^+, converting it to NADH and leading to reductive stress[70]. Subsequently, excess electrons bond with molecular oxygen to produce ROS, which act as potent signalling molecules.

With ageing, the balance of oxygen supply and demand shifts. The endothelial lining of blood vessels gradually loses its responsiveness to stimuli, such as shear stress and cytokines, leading to decreased production of NO. This decline is exacerbated by oxidative stress, which promotes uncoupling of NO synthase and oxidation of the haem group within soluble guanylyl cyclase, thereby impairing NO signalling[71,72]. In addition, arteries become stiffer due to collagen accumulation and undergo a reduction in elastic fibres, which together impair their ability to expand and contract effectively, particularly in the context of hypertension. Supporting this perspective, evidence suggests that older individuals experience impaired peripheral oxygen utilization during exercise[73]. This phenomenon cannot be solely attributed to a decline in cardiac output, and is probably related to reduced tissue perfusion or diminished mitochondrial oxidative phosphorylation.

Ageing reduces the sensitivity and responsiveness of VSMCs to neurohormonal signals, such as catecholamines and endothelins[74]. In addition, tissue fibrosis can disrupt normal perfusion by altering the architecture of the vascular network, leading to decreased capillary density, reduced lumen diameter, and diminished flexibility and compliance[75]. These changes collectively increase vascular resistance. As a result, the potential for tissues to adapt to fluctuations in oxygen demand or metabolic activity is impaired, leading to chronic or episodic hypoxia. The risk of chronic hypoxia is further heightened by inflammation, particularly in tissues infiltrated by immune cells. Obesity and chronic multimorbidity can elevate metabolic demands, exacerbating the risk of tissue hypoxia.

Despite the importance of both mild hypoperfusion and chronic hypoxia in both health and disease, research on their physiological consequences has been limited. Hypoperfusion is difficult to assess, as clinical signs typically only appear when the condition is severe. For example, circulating lactate levels can serve as a biomarker of perfusion, but lactate production requires a substantial metabolic shift and thus is only detectable in the most severe cases[76]. Moreover, existing technologies for the detection of hypoperfusion, such as sidestream dark-field imaging techniques[77], are neither sensitive nor user friendly.

Learning from PAD. Valuable insights into tissue hypoperfusion can be gleaned from studying the effects of persistent low-grade hypoxia and ischaemia–reperfusion in patients with peripheral artery disease (PAD). PAD is characterized by narrowed peripheral arteries, often due

Fig. 4 | Protein biomarkers of cardiovascular disease. The heat map shows proteins (rows) associated with cardiovascular-related outcomes in at least six studies (columns). For each gene and protein product, the tissues listed in parentheses represent those in which mRNA was identified in the GTEx repository[114] to be highly and selectively overexpressed (*Z* score >2)[117]. Labels along the *x* axis list the first author, year and outcomes of each study, followed by the reference number (for more details, see Supplementary Table 1). The biological pathways associated with each protein have been manually curated based on a comprehensive literature review. Proteins with a known direct role in immunity are indicated by a red dot, whereas those with an indirect role in inflammation are indicated by a yellow dot. The 'pathway' column provides information on the proteins that have been previously identified as senescence-associated secretory phenotype (SASP) members based on a literature search of the SASP atlas and other evidence[108–113]. [a]Pseudogenes are non-functional segments of DNA that resemble functional genes, but do not produce proteins. Some sequences originally coded as pseudogenes are actually expressed and translated. Database overlap and sequence redundancy can cause ambiguous peptide mapping. In addition, biological and technical noise can create apparent pseudogene hits. We decided to include basic transcription factor 3 pseudogene 11 (BTF3P11) and leukocyte immunoglobulin-like receptor pseudogene 2 (LILRP2) in our analysis because they are reported in the multiple papers as associated with cardiovascular disease (CVD). AAA, abdominal aortic aneurysm; ACE2, angiotensin-converting enzyme 2, ADGRG2, adhesion G protein coupled receptor G2; ADM, adrenomedullin; AF, atrial fibrillation; AMBP, protein AMBP; ANGPT2, angiopoietin-2; AS, aortic stenosis; ASCVD, atherosclerotic cardiovascular disease; BCAN, brevican core protein; BST2, bone marrow stromal antigen 2; CAD, coronary artery disease; CD302, CD302 antigen; CEACAM5, cell adhesion molecule CEACAM5; CM, cardiomyopathy; COL18A1, collagen alpha-1(XVIII) chain; CST3, cystatin-C; CXCL, C-X-C motif chemokine; DC, dilated cardiomyopathy; EDA2R, tumour necrosis factor receptor superfamily member 27 (also known as ectodysplasin-A2 receptor); ECM, extracellular matrix; EFEMP1, EGF-containing fibulin-like extracellular matrix protein 1; EGFR, epidermal growth factor receptor; FGF23, fibroblast growth factor 23; GDF15, growth/differentiation factor 15; HAVCR1, hepatitis A virus cellular receptor 1; HF, heart failure; HT, hypertension; ICAM1, intercellular adhesion molecule 1; IGFBP, insulin-like growth factor-binding protein; IHD, ischaemic heart disease; KLK4, kallikrein 4; LTBP2, latent-transforming growth factor β-binding protein 2; MACE, major adverse cardiovascular event; MI, myocardial infarction; MMP, matrix metalloproteinase; N/A, not available; NEFL, neurofilament light polypeptide; NMVD, non-rheumatic mitral valve disorders; NPPB, natriuretic peptide B; PCSK9, proprotein convertase subtilisin/kexin type 9; PLAT, tissue-type plasminogen activator; PLAUR, urokinase plasminogen activator surface receptor; PPHT, primary pulmonary hypertension; REG1A, lithostathine-1-α (also known as regenerating protein Iα); REN, renin; SPHT, secondary pulmonary hypertension; SPON1, spondin 1; SPP1, secreted phosphoprotein 1 (also known as osteopontin); TFF3, trefoil factor 3; TIMP, metalloproteinase inhibitor; TNFRSF, tumour necrosis factor receptor superfamily; TNNI3, troponin I, cardiac muscle; VWC2, brorin (also known as von Willebrand factor C domain-containing protein 2); WFDC2, WAP four-disulfide core domain protein 2.

to atherosclerosis, which commonly leads to limb pain during physical activity that eases with rest. The clinical diagnosis of PAD typically involves measuring the ankle–brachial index (ABI), a non-invasive test that compares blood pressure in the ankle to that in the arm. An ABI value of ≤0.9 is diagnostic for PAD[78]. Interestingly, low ABI values within the normal range (0.9–1.4) have been associated with reduced mitochondrial oxidative phosphorylation, which is likely to stem from chronic hypoxia[79]. In PAD, chronic hypoxia and repeated episodes of ischaemia–reperfusion can induce myopathy that is characterized by mitochondrial dysfunction and the activation of various resilience response mechanisms. These mechanisms include angiogenesis, fibrosis, the integrated stress response (ISR) and UPR (discussed in the later section on the UPR and ISR in tissue hypoperfusion), as well as the upregulation of erythropoietin and a metabolic shift towards glycolysis. These events can trigger an inflammatory response that can negatively affect tissue function and health.

Studies in animal models of PAD and patients with PAD have suggested that hypoperfusion disrupts the flow of electrons through mitochondrial respiratory complexes, leading to impaired oxidative phosphorylation and ATP production. This disruption can also result in electron leakage generating ROS, which activate pro-inflammatory transcription factors, such as NF-κB. In turn, these proteins increase the expression of pro-inflammatory cytokines, including IL-1β, IL-6 and TNF[6]. Furthermore, ROS can damage mitochondrial membranes, proteins and DNA, activating additional inflammatory signalling pathways.

Stressed mitochondria release DAMPs, such as ATP, cytochrome *c*, oxidized cardiolipin and mtDNA, into the cytoplasm or extracellular space[6]. These DAMPs can stimulate immune responses by activating pattern recognition receptors on immune cells, particularly Toll-like receptors, leading to further inflammation. mtDNA can also activate the NACHT, LRR and PYD domains-containing protein 3 (NLRP3) inflammasome, a multiprotein complex that is crucial for immune responses, resulting in the production of pro-inflammatory cytokines such as IL-1β and IL-18 (ref. 6).

mtDNA is structurally similar to bacterial and viral RNA and, therefore, its presence in the cytosol signals danger to the immune system[80]. mtDNA is detected by the cytosolic second messenger enzyme cyclic GMP–AMP synthase (cGAS), which promotes the dimerization of stimulator of interferon genes protein (STING), leading to a conformational change that allows STING to transit from the endoplasmic reticulum to the Golgi apparatus[80]. STING subsequently interacts with kinases, such as serine/threonine-protein kinase TBK1, facilitating the phosphorylation of the transcription factor interferon regulatory factor 3 (IRF3). IRF3 then translocate to the nucleus to induce the transcription of various type I interferons (for example, IFNβ) and other genes involved in the inflammatory response.

Thus, mitochondrial stress activates three critical inflammatory pathways — NF-κB, the NLRP3 inflammasome and the cGAS–STING pathway — culminating in a robust inflammatory response. Indeed, studies in humans have found that skeletal muscle mitochondrial oxidative phosphorylation declines with ageing and is associated with a pro-inflammatory state[81]. Severe mitochondrial stress can lead to the opening of the mitochondrial membrane permeability transition pore, resulting in the release of apoptotic factors and consequent cell death, which further contributes to inflammation[82]. Stressed mitochondria should, theoretically, be eliminated through several forms of mitophagy[20]. However, this process can become dysfunctional with ageing and, in hypoperfused tissues, non-functional fragments of mitochondria often persist. In addition, hypoxia increases the stabilization of hypoxia-inducible factor 1 (HIF1), the master regulator of the hypoxic response, which inhibits mitochondrial biogenesis and promotes fission, resulting in mitochondrial fragmentation. Increased levels of HIF1 also promotes the transcription of erythropoietin, which stimulates the production of erythrocytes to enhance oxygen transport. Although ageing is associated with progressively increasing levels of erythropoietin, such compensation is not fully effective because circulating haemoglobin declines with age, possibly because of bone marrow insufficiency[83].

The UPR and ISR in tissue hypoperfusion. During hypoxia, the accumulation of misfolded proteins triggers a proteostatic response, prominently featuring the UPR[84]. This response inhibits the translation of most proteins while increasing the production of molecular chaperones that assist in refolding or degrading misfolded proteins. In addition, the UPR activates inflammatory mediators, including cyclic AMP-dependent transcription factor ATF6, and the NF-κB and c-Jun N-terminal kinase pathways, increasing the expression of pro-inflammatory cytokines, such as IL-6 and TNF[85]. Consequently, persistent activation of the endoplasmic reticulum stress response can lead to sustained inflammation. In cases of severe, prolonged hypoxia, the UPR can induce cellular senescence, resulting in the production of the SASP, which further drives chronic inflammation and tissue dysfunction[86].

The UPR is complemented by the ISR[86,87], which is activated by a family of kinases — chiefly eukaryotic translation initiation factor 2α (eIF2α) kinase 3 (also known as PKR-like endoplasmic reticulum kinase) — that responds to endoplasmic reticulum stress; eIF2α kinase GCN2, which detects uncharged transfer RNAs as indicators of nutrient or energy deprivation during hypoxia and reductive stress[70]; and interferon-induced, double-stranded RNA-activated protein kinase, which responds to viral infections or general cellular stress. Activation of these kinases leads to the phosphorylation of eIF2α, a key component of the eukaryotic translation initiation complex that facilitates the binding of the initiator transfer RNA (tRNA) to the ribosome. Phosphorylation of eIF2α alters its conformation, reducing the availability of the active form of eIF2 (the eIF2–GTP–tRNA complex). As a result, the binding of initiator tRNA to the ribosome is impaired, effectively blocking the formation of the translation initiation complex. The phosphorylation of eIF2α also allows the translation of stress-responsive mRNAs, including those encoding transcription factors, such as cyclic AMP-dependent transcription factor ATF4 that promotes the expression of target genes encoding stress response biomarkers (such as GDF15, IL-6 and TNF) and other inflammatory mediators. The ISR also influences immune signalling pathways, including the NF-κB and tyrosine-protein kinase JAK–signal transducer and transcription activator (JAK–STAT) pathways, thereby amplifying inflammatory responses.

Although the UPR and ISR have been studied independently, they share similarities and contribute to overlapping resilience mechanisms. When stress remains unresolved, owing to either prolonged duration or overt intensity, these protective mechanisms can trigger programmed cell death to safeguard the organism. These stress response pathways have been implicated in various CVDs and consistently elicit immune responses[88].

Chronic hypoperfusion as a cause of inflammation. As we have discussed, maladaptive age-related derangement of the mechanisms that maintain adequate tissue perfusion can cause mitochondrial dysfunction, proteostatic stress and inflammation. These events trigger a vicious cycle that impairs tissue perfusion, causes hypoxia and perpetuates the inflammatory response. Studies have shown that DNA methylation sites that change systematically with ageing are located in proximity to transcription factor-binding sites for hypoxia mediators[89]. Glucagon-like peptide 1 receptor (GLP1R) agonists have been shown to effectively prevent several cardiovascular conditions and reduce chronic inflammation independent of their effect on insulin release and appetite[90]. Evidence suggests that the beneficial effects of GLP1R agonists on CVD and inflammatory status are partly mediated through improvements in endothelial function and the microcirculation in the gut, heart, liver, skeletal muscle and brain[91–93]. The effects are specifically enabled by increased NO production and eNOS phosphorylation through activation of the 5′-AMP-activated protein kinase–phosphoinositide 3-kinase–serine/threonine-protein kinase Akt–eNOS pathway via a GLP1R–cyclic AMP-dependent mechanism[94]. In fact, the GLP1R agonist, liraglutide, has been shown to enhance dilation of microvessels by both NO-dependent (such as increased circulating insulin) and NO-independent mechanisms[95]. Overall, there is a growing body of evidence indicating that physiological dysfunction of the microcirculation can accelerate macromolecular damage, which causes inflammation and contributes to CVD pathology.

Inflammation as a biomarker of CVD risk

In the penultimate section of this Review, we examine the growing body of literature on circulating proteins associated with the risk of developing CVD. The analysis of thousands of proteins from small volumes of plasma, serum and other biological matrices is now possible. Coupled with an unbiased methodological approach, this technology has expanded our understanding of the complex mechanisms linked to CVD pathology. We conducted a literature search to identify manuscripts that detailed unbiased, prospective studies using an 'omics' approach to protein analysis and with one or more cardiovascular conditions as outcomes[96–107] (Supplementary Table 1). The heat map in Fig. 4 highlights proteins (rows) found to be significantly associated with cardiovascular outcomes in at least six analyses (columns) reported in those studies. Supplementary Table 2 provides further details on the 51 proteins uncovered so far. Notably, most of the circulating proteins identified as CVD biomarkers have been mentioned in the context of atherosclerosis and chronic tissue hypoperfusion in this Review. The strong connection between mechanistic studies of CVD pathogenesis and circulating biomarkers suggests that changes in these biomarkers could serve as surrogate indicators of CVD mechanisms and pathology.

Prominent proteins related to immune activation include protein AMBP, CD302 antigen, CXCL9, CXCL17, hepatitis A virus cellular receptor 1, ICAM1, IL-6, urokinase plasminogen activator surface receptor, tumour necrosis factor receptor superfamily (TNFRSF) member 1B, TNFRSF member 6B, TNFRSF member 9, TNFRSF member 10A and WAP four-disulfide core domain protein 2 (Fig. 4). Some receptors, such as the epidermal growth factor receptor and TNFRSF member 10A, connect inflammation with vital processes including survival, proliferation and apoptosis, all of which affect cardiovascular cell behaviour and response to injury. However, inflammation is not the sole pathway of note. For example, proteins such as angiotensin-converting enzyme 2, adrenomedullin, natriuretic peptide B and renin are important in cardiovascular homeostasis, regulating blood pressure, heart function and fluid balance. In addition, although GDF15 is an immunomodulator, it is also thought to be a multifaceted stress-response biomarker, not limited to the modulation of inflammation. Many other proteins, including brevican core protein, EGF-containing fibulin-like extracellular matrix protein 1, latent-transforming growth factor β-binding protein 2, MMP7, MMP12, metalloproteinase inhibitor 1 (TIMP1) and TIMP4, are crucial for ECM remodelling, a process essential for maintaining structural integrity and responding to injury in cardiovascular tissues. Disruptions in ECM maintenance can lead to fibrosis and atherosclerosis. As discussed in the sections on microvascular tissue hypoperfusion, inflammation, fibrosis and changes in the intercellular matrix can contribute to hypoxia, triggering a series of events that initiate and

Table 1 | Interventions targeting ageing-related mechanisms upstream of inflammation

Intervention	Mechanism of action (upstream of inflammation)	Representative clinical studies (reference)
Dietary inorganic nitrate (beetroot juice)	Restores NO bioavailability via the nitrate–nitrite–NO pathway; enhances endothelial function and tissue perfusion	118
Tetrahydrobiopterin (BH4; sapropterin dihydrochloride)	eNOS cofactor; 'recouples' eNOS to favour NO over superoxide, improving endothelial signalling	119
PDE5 inhibition (sildenafil)	Inhibits cGMP breakdown, leading to augmented NO–cGMP signalling, which improves endothelial and microvascular perfusion	120
Dasatinib plus quercetin (senolytic[a])	Transiently inhibits SCAP pathways and results in the apoptosis of senescent cells, which reduces the release of SASP cytokines	121
Metformin (senomorphic[b])	Activates AMPK and inhibits NF-κB and SASP, which improves endothelial function independent of glycaemia	122
mTOR inhibition (everolimus/RTB101)	Dampens mTORC1 signalling, resulting in a reduction in cellular senescence-like phenotype and pro-inflammatory signalling	123
Mitoquinone mesylate (MitoQ; mitochondria-targeted antioxidant)	Localizes to mitochondria, reduces mtROS, restores endothelial NO bioavailability and reduces oxidative stress	124
Nicotinamide riboside (NAD^+ precursor)	Boosts NAD^+ levels, improving mitochondrial function, vascular stiffness and energy metabolism	125
Coenzyme Q10 (ubiquinone/ubiquinol)	An electron transport chain cofactor that improves myocardial bioenergetics, and reduces oxidative stress	126
Elamipretide (SS-31)	Binds cardiolipin and stabilizes the inner mitochondrial membrane, improving mitochondrial coupling and ATP production	127
Urolithin A (Mitopure)	Induces mitophagy and mitochondrial biogenesis, enhances mitochondrial quality control, and reduces mtROS and inflammatory signalling	128–130

[a]Selectively eliminates senescent cells. [b]Modifies the pro-inflammatory activity of senescent cells. AMPK, 5′-AMP-activated protein kinase; eNOS, endothelial nitric oxide synthase; mtROS, mitochondrial reactive oxygen species; mTOR, serine/threonine-protein kinase mTOR; mTORC1, mechanistic target of rapamycin complex 1; NF-κB, nuclear factor-κB; NO, nitric oxide; PDE5, cGMP-specific phosphodiesterase type 5; SCAP, sterol regulatory element-binding protein cleavage-activating protein; SASP, senescence-associated secretory phenotype.

exacerbate CVD. Moreover, proteins, such as angiopoietin-2, FGF23 and secreted phosphoprotein 1 (also known as osteopontin) are involved in in vascular remodelling and angiogenesis, processes that are integral to ischaemic CVD.

Growth factor regulators, such as insulin-like growth factor-binding protein 1 (IGFBP1), IGFBP2 and IGFBP7, also found in our analysis, influence cellular processes related to growth and repair, especially under stress conditions. Proteolytic enzymes, such as cystatin-C and kallikrein 4, have important roles in proteostatic mechanisms. Notably, the only protein identified that regulates cholesterol is PCSK9, which binds to LDL receptors on liver cells, promoting their degradation.

The diverse biological functions of these proteins reveal potential therapeutic targets for modulating inflammatory responses and disease mechanisms. If this belief holds true, cytokine signalling could serve as a marker for identifying effective interventions to prevent and treat CVD. Strikingly, 28 of the 50 plasma proteins identified as predictors of cardiovascular outcomes have been previously identified as SASP members in preclinical studies of cellular senescence, and seven others are receptors of SASP members[108–113] (Fig. 4). These findings suggest a strong link between CVD pathology and cellular senescence, a hallmark of ageing, raising the possibility that many identified pro-inflammatory mediators might share a common origin.

By examining the GTEx database[114], we found that the genes encoding these proteins are preferentially expressed in a variety of tissues beyond the cardiovascular system (Fig. 4), most notably the kidney, liver, lungs and salivary glands. Although these data should be interpreted cautiously, they suggest that CVD is closely linked to pathological processes that originate outside the cardiovascular system. Research published in the past 2 years on organ-specific ageing clocks inferred from the plasma proteome supports this hypothesis[115,116].

Conclusions

The relationship between inflammation and CVD is well supported by preclinical, clinical and epidemiological evidence. Emerging findings indicate that stress signalling in both immune and non-immune cells can be initiated by the accumulation of molecular damage during the early stages of atherosclerosis. We hypothesize that chronic tissue hypoperfusion, common during ageing and metabolic disease, also contributes to inflammatory activation. Consequently, the constellation of cytokines released during inflammation should be viewed as biomarkers of stress-response mechanisms triggered by accumulated molecular and cellular damage or by energetic stress that, over time, can exacerbate CVD pathology.

Examples of interventions that target upstream mechanisms of inflammation are summarized in Table 1, and include agents that enhance tissue perfusion and NO signalling, selectively eliminate or modulate senescent cells to suppress the SASP or improve mitochondrial respiratory capacity and quality control through the activation of mitophagy or a reduction in mitochondrial oxidative stress. Collectively, these strategies address fundamental processes related to damage accumulation that promote vascular inflammation and cardiovascular pathology during ageing. As our understanding of these and other age-related mechanisms deepens, novel treatments aimed directly at the biological damage that initiates inflammation may offer new avenues for the prevention and management of CVD.

Glossary

Biomarkers
Measurable indicators of biological and pathogenic processes or therapeutic efficacy.

Cellular senescence
A state of stable cell cycle arrest caused by sublethal DNA damage and stress.

Entropy
The measure of a system's thermal energy per unit temperature that is unavailable for doing useful work.

Immunosenescence
Innate and adaptive immune dysregulation that occurs with ageing.

Integrated stress response
The signalling pathway activated in response to various forms of stress, including nutrient and oxygen deprivation, oxidative stress and viral infections.

Macromolecular damage
The accumulation of structural and chemical alterations in DNA, proteins, lipids and polysaccharides within cells.

Perfusion
The passing of a fluid through spaces.

Proteostasis or proteostatic
Protein homeostasis or homeostatic state.

Resilience response
The capacity of an organism to adapt or recover from stress. The response involves molecular pathways that detect stress, promote survival and enable repair.

Senescence-associated secretory phenotype
The secretion of bioactive molecules associated with damage-induced cellular senescence.

Stochastic
Occurring by chance or random variables.

Thermodynamics
The relationships between heat, work, temperature and energy.

Unfolded protein response
The signalling pathway responsible for the fidelity of protein folding in the lumen of the endoplasmic reticulum.

Data availability

Datasets and the custom R source code used in the proteomic meta-analysis are available on the Open Science Framework repository[117].

Published online: 22 January 2026

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Acknowledgements
The authors acknowledge the support of the Intramural Research Program of the National Institute on Aging, National Institutes of Health, Baltimore, MD, USA.

Author contributions
All the authors contributed substantially to all aspects of the article.

Competing interests
The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41569-026-01253-3.

Peer review information *Nature Reviews Cardiology* thanks Mika Kivimaki, Esther Lutgens, Andrew Steptoe and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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